GG-S-921b
AUGUST 7, 1958
SUPERSEDING
Fed. Spec. GG–S–921a
June 14, 1940

FEDERAL SPECIFICATION

SYRINGE, LUER, (ALL-GLASS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—This specification covers all-glass, Luer-slip-type syringes, suitable for use in medical and surgical procedures.

1.2 **Classification.**

1.2.1 *Types and sizes.*—Syringes shall be of the following types and sizes, as specified (see 6.1) :

Type I.—General purpose syringe—small Luer tip.
- Size 1.—2 cubic centimeters.
- Size 2.—5 cubic centimeters.
- Size 3.—10 cubic centimeters.
- Size 4.—20 cubic centimeters.

Type II.—General purpose syringe—large Luer tip—with metal needle adapter.
- Size 1.—30 cubic centimeters.
- Size 2.—50 cubic centimeters.
- Size 3.—100 cubic centimeters.

Type III.—Special purpose syringe.
- Size 1.—0.25 cubic centimeter for Tuberculin.
- Size 2.—0.50 cubic centimeter for Mallein or Tuberculin.
- Size 3.—1 cubic centimeter for Tuberculin.
- Size 4.—1 cubic centimeter for 20-40-unit Insulin—short model.
- Size 5.—1 cubic centimeter for 20-40-unit Insulin—long model.
- Size 6.—1 cubic centimeter for 40-80-unit Insulin—long model.
- Size 7.—1 cubic centimeter for Insulin of all concentrations (universal scale) —long model.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.**—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:
- PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
- PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:
- Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–4229 — Boxes; Paperboard, Metal-Stayed.

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling, Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Chemical Society Publication:

Reagent Chemicals.

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth Street, N. W., Washington 6, D. C.)

U. S. Pharmacopoeial Convention Publication:

Pharmacopoeia of the United States.

(Application for copies should be addressed to the Mack Printing Company, Easton, Pa.)

American Standards Association Publication:

ASA Z70.1–1955 (UDC615.47) American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American Standards Association, Incorporated, 70 East Forty-fifth Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Barrel and plunger.* — Barrel and plunger shall be fabricated of high-grade, heat-resistant, corrosion-resistant borosilicate glass throughout. The glass shall have a density of not more than 2.4, and shall withstand the tests specified in section 4.

3.1.2 *Metal needle adapter.*—Needle adapter for type II syringes shall be fabricated of either corrosion-resistant metal, or brass, nickel, or chromium over nickel plated, suitable for the purpose.

3.2 Construction.—Syringes shall consist of a barrel and plunger, as hereinafter specified. Unless otherwise specified (see 6.1), type II syringes shall be equipped with a needle adapter.

3.2.1 *Barrel.*—Barrel shall be a tube closed at one end, except for an effluent, and open at the other end. Barrel shall have attached thereon at the closed end, a Luer tip for attaching a syringe needle. The wall of the open end shall be thickened, and extended radially outward, to form a flange to prevent slipping of the barrel through the fingers. The inside surface of the barrel shall be finished to a very fine, uniform, ground surface, or a smooth clear unground surface, free from pits, grind marks, or open-air lines. There shall be no high spots along the periphery where the bottom joins the interior wall of the barrel. The distal end of the plunger, when fully inserted, shall mate evenly with the distal end of the barrel.

3.2.1.1 *Flange.*—The form and dimensions of the flange shall be appropriate to the syringe and shall extend outward from the wall, sufficient to give ample fingerhold, and lengthwise of the barrel sufficient for strength and durability. Opposite segments of the flange shall be removed to provide straight parallel sides of ample length to prevent rolling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

3.2.1.2 *Tip.* — Unless otherwise specified (see 6.1), the tip shall be formed concentric to the axis of the barrel. The tip shall be two sizes as hereinafter specified, and shall have a needle cone accurately ground to a uniform taper of 0.060 inch per inch of length. Under a low power (4 to 8 X) magnification, the cone shall show a fine, smooth-ground surface relatively free from pits, open air lines, and grind marks, and shall make a secure, watertight union with a needle hub or needle adapter having a corresponding taper. The tip shall be joined to the barrel with a generous fillet in such a manner as to form a smooth, strong union with the barrel. The effluent through the tip shall be concentric therewith and shall be of the diameter indicated in tables I through III.

3.2.1.2.1 *Tip for types I and III syringes.*— Tip for syringes 2 cc. and smaller, shall measure 0.340 ± 0.063 inch in overall length. Tip for syringes, 5, 10, and 20 cc., shall measure 0.460 ± 0.125 inch in overall length. The ground tapered portion of the tip (needle cone) shall be as shown in figure 2 (male end) of American Standards Association (ASA) Standard Z70.1–1955. When tested as specified in 4.3.3.1, the ground tapered portion of the tip shall fit the gage within the "GO" —"NO GO" limits and there shall be no movement of the gage.

3.2.1.2.2 *Tip for type II syringe.*—Tip for type II syringe shall measure 0.700 ± 0.125 inch in over-all length. The length of the ground taper shall be not less than 0.250 inch. The tip shall accurately fit the female taper gage specified in 4.3.3.2 without movement of the gage.

3.2.1.3 *Graduations and other markings.*— The outside surface of the barrel shall be marked with the graduation scale and other markings required for the particular type of syringe as hereinafter specified, the capacity of the syringe, serial number (corresponding to serial number on plunger), the name or registered trade-mark of the manufacturer or contractor, and the words "Property of the U.S.". Graduations and other markings for type III, sizes 4 and 5 syringes shall be in red color except that serial number may be in amber color. The syringes shall also be marked "Use U–40 only" in addition to the other required markings. Graduations and other markings for type III, size 6 syringes shall be in green color, except that serial number may be in amber color. This syringe shall also be marked "Use U–80 only" in addition to the other required markings. All of these markings shall be applied to the barrel in such a manner that they are in line with the flattened side of the flange. Main graduation marks shall be at least 33⅓ percent longer than the subgraduation marks, and the "0" mark of the scale shall register precisely with the extreme distal end of the plunger, or with the dark colored mark on the bevel, when the plunger is pushed to the bottom of the barrel. All markings shall be clearly legible and either fused in, fused on, or etched and fused. The marking shall withstand the permanency of marking test specified in 4.3.7.

3.2.2 *Plunger.*—Plunger for syringes of 5 cc. capacity and over shall be hollow; for syringes under 5 cc. capacity plunger may be hollow or solid. The distal end of the plunger shall be ground flat, or formed (unground) flat, and at right angles to the axis, and its edge shall be slightly beveled. A permanent mark of dark color shall appear on the beveled edge, or the plunger may be of dark colored glass throughout its length, or for part of its length from the distal end. The distal end of the plunger shall be clearly visible and accurately indicated through the barrel wall. The proximal end of the plunger shall extend well out of the barrel and be shaped to form a button-type knob. The sides of the plunger shall show under low power (4 to 8 X) magnification, a uniform, smooth-ground surface, free from pits, grind marks, or open-air lines, and shall accurately fit its individual barrel so as to make the syringe successfully withstand the applicable leak tests, specified in section 4, and yet move freely throughout its excursion in the barrel. Plunger shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

permanently marked near the proximal end with a serial number, corresponding to that on the barrel. Pigment shall correspond to that used for barrel markings. This marking shall withstand the permanency of marking test specified in 4.3.7.

3.2.3 *Metal needle adapter.*—Metal needle adapter for type II syringes, when furnished as specified, shall consist of a cylinder fabricated of material specified in 3.1.2 and shall measure ¾ ± 1/16 inch in length and 5/16 ± 1/32 inch in diameter, having in its proximal end a Luer cavity with a uniform taper of 0.060 inch per inch of taper, finished to make a secure fluid-tight slip connection with the larger Luer tip of the type II syringe, and the other end finished with a standard male Luer taper the same size as the needle cone required for types I and III syringes. Adapter shall be drilled through on its axis to provide an effluent, corresponding to that of the syringe. When tested as specified in 4.3.3.1, the male Luer taper portion (needle cone) shall fit the gage within the "GO"—"NO GO" limits and there shall be no movement of the gage.

3.3 **Dimensions.**

3.3.1 *Type I syringes.*—Dimensional details of barrels for type I syringes, including graduation markings, shall be as indicated in table I.

3.3.1.1 *Graduation scale.* — Unless otherwise specified (see 6.1), graduation scale for type I syringes shall be in cubic centimeters, except that size 1 syringe (2 cc.) shall have a double scale, one in cubic centimeters with main graduations at 0.5 cc. intervals marked ½, 1, ½, and 2 cc. successively, with subgraduations at 0.1 cc. intervals unmarked; the other to be a minim scale with main graduations at 5 minim intervals, marked 10, 20, and 30 successively, with subgraduations at 1 minim intervals unmarked.

3.3.2 *Type II syringes.*—Dimensional details of barrels for type II syringes, including

TABLE I.—*Dimensional details of barrel for type I syringes*

Size	Capacity ± 5%	Length		Diameter	Wall thickness	Diameter effluent	Graduations	
		Overall, not including tip	Portion ungraduated	Outside	Minimum		Sub.	Main
	cc.	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*cc.*	*cc.*
1	2	2.1–2.25	0.55–1.17	0.4–0.5	0.035	0.030–0.050	*	*
2	5	2.7–3.0	0.6–1.6	0.5–0.7	0.042	0.030–0.050	0.2	1.0
3	10	3.6–3.85	0.75–1.9	0.6–0.8	0.045	0.030–0.050	0.2	1.0
4	20	4.2–4.4	1.1–2.2	0.8–1.0	0.050	0.030–0.060	1.0	5.0

* See 3.3.1.1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:53Z
Check the source to verify that this is the current version before use.

graduation markings, shall be as indicated in table II.

3.3.2.1 *Metal needle adapter.* — Unless otherwise specified (see 6.1), a metal needle adapter, as specified in 3.2.3, shall be furnished with type II syringe.

3.3.3 *Type III syringes.*—Dimensional details of barrels for type III syringes, including graduation markings, shall be as indicated in table III.

3.3.3.1 *Graduation scale.*

3.3.3.1.1 *Size 1.*—Size 1 syringe (0.25 cc.) shall have a double scale, one in cubic centimeters with main graduations at 0.05 cc. intervals marked 5, 10, 15, 20, and ¼ cc. successively, with subgraduations at 0.01 cc. intervals unmarked; the other to be a minim scale with main graduations at 1 minim intervals marked 1, 2, 3, and 4 successively, with subgraduations at 0.5 minim intervals unmarked.

3.3.3.1.2 *Size 2.*—Size 2 syringe (0.5 cc.) shall have a double scale, one in cubic centimeters with main graduations at 0.10 cc. intervals marked 10, 20, 30, 40, and ½ cc. successively, with subgraduations at 0.01 cc. intervals unmarked; the other to be a minim scale with main graduations at 4 and 8 successively, with subgraduations at 0.5 minim intervals unmarked.

3.3.3.1.3 *Size 3.*—Size 3 syringe (1 cc. for Tuberculin) shall have a double scale, one in cubic centimeters with main graduations at 0.20 cc. intervals marked 20, 40, 60, 80, and 1 cc. successively, with subgraduations at 0.01 cc. intervals unmarked; the other to be a minim scale with main graduations at 4 minim intervals marked 4, 8, 12, and 16 successively, with subgraduations at 0.5 minim intervals unmarked.

3.3.3.1.4 *Sizes 4, 5, and 6.*—Sizes 4, 5, and 6 (Insulin syringes) shall have a single scale. The 40-unit scale shall have main graduations at 5-unit intervals with the alternate graduations marked 10, 20, 30 and 40, and with subgraduations at 1-unit intervals unmarked. The 80-unit scale shall have main graduations at 5-unit intervals marked 10, 20, 30, 40, 50, 60, 70, and 80 with subgraduations at 1-unit intervals unmarked. As an al-

TABLE II.—*Dimensional details of barrel for type II syringes*

Size	Capacity ± 5%	Length		Diameter	Wall thickness	Diameter effluent	Graduations	
		Overall, not including tip	Portion ungraduated	Outside	Minimum		Sub.	Main
	cc.	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*cc.*	*cc.*
1	30	4.65–4.95	1.2–2.25	0.9–1.2	0.054	0.065–0.090	1	5
2	50	5.0–5.3	1.2–2.6	1.1–1.4	0.065	0.065–0.090	5	10
3	100	6.1–6.5	1.5–2.85	1.4–1.7	0.075	0.065–0.090	10	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

TABLE III.—*Dimensional details of barrel for type III syringes*

Size	Capacity ± 5%	Length		Diameter	Wall thickness	Diameter effluent	Graduations (See paragraph)
		Overall, not including tip	Portion ungraduated	Outside	Minimum		
	cc.	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	
1	0.25	2.3–2.5	0.5–1.37	0.2–0.32	0.035	0.030–0.050	3.3.3.1.1
2	0.5	2.7–2.9	0.6–1.6	0.2–0.35	0.035	0.030–0.050	3.3.3.1.2
3	1	3.35–3.6	0.7–1.65	0.3–0.4	0.035	0.030–0.050	3.3.3.1.3
4	1	2.1–2.3	0.5–1.2	0.3–0.4	0.035	0.030–0.050	3.3.3.1.4
5	1	3.4–3.6	0.7–1.6	0.3–0.4	0.035	0.030–0.050	3.3.3.1.4
6	1	3.4–3.6	0.7–1.6	0.3–0.4	0.035	0.030–0.050	3.3.3.1.4
7	1	3.4–3.6	0.7–1.6	0.3–0.4	0.035	0.030–0.050	3.3.3.1.5

ternate, the 80-unit scale shall have main graduations at 10-unit intervals marked 20, 40, 60, and 80 with subgraduations at 2-unit intervals unmarked. Each scale shall bear the word "Units" and the capacity 1 cc. clearly marked at the proximal end of the barrel.

3.3.3.1.5 *Size 7.*—Size 7 syringe (universal scale) for Insulin of all concentrations shall be legibly marked with the capacity near the proximal end of the syringe. The word "Unit" or "Units" shall not appear on the syringe. Syringe shall have a single cubic centimeter scale with graduations at 1/10 cc. intervals marked 1, 2, 3, 4, 5, 6, 7, 8, 9, and 1 cc. successively.

3.4 Fluid tightness.

3.4.1 *Fluid tightness between plunger and barrel.*—The syringe shall be tested for fluid tightness between the plunger and barrel, as specified in 4.3.1.1. During the first test, there shall not be enough water to pass the plunger to produce a falling drop. After each successive test, any water that may have passed the plunger shall be wiped away and the next test applied, during which there shall not be enough water to pass the plunger to produce a falling drop.

3.4.2 *Fluid tightness, syringe tip to hypodermic needle hub.*—The syringe tip shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

tested for fluid tightness between the tip and hypodermic needle hub, as specified in 4.3.1.2. When pressure is applied for 10 seconds, there shall be no leakage exceeding one falling drop between the syringe tip and the hub of the gage.

3.4.3 *Fluid tightness, needle adapter to hypodermic needle hub.*—The needle adapter tip shall be tested for fluid tightness between the adapter tip and hypodermic needle hub, as specified in 4.3.1.3. When pressure is applied for 10 seconds, there shall be no leakage exceeding one falling drop between the needle adapter tip and the hub of the gage.

3.4.4 *Fluid tightness, needle adapter to syringe tip.*—The needle adapter shall be tested for fluid tightness between the needle adapter and syringe tip as specified in 4.3.1.3. When pressure is applied for 10 seconds, there shall be no leakage exceeding one falling drop beteen the syringe tip and the needle adapter.

3.5 **Alkalinity.**—Syringes shall be tested for alkalinity as specified in 4.3.2. The volume of 0.02 Normal (N) sulphuric acid required to neutralize the pink color, which may be observed after the solution has cooled, shall not exceed the amounts specified in table IV.

TABLE IV.

Type	Size	Capacity	Maximum volume of 0.02N acid to neutralize alkali (per syringe)
		cc.	cc.
I	1	2	0.02
	2	5	0.04
	3	10	0.05
	4	20	0.08
II	1	30	0.09
	2	50	0.16
	3	100	0.20
III	All sizes	---	0.02

3.6 **Resistance to autoclaving.**—The syringes shall be tested for resistance to autoclaving as specified in 4.3.4. At the time the test is completed (while syringes are still hot), the plunger shall be freely movable in the barrel.

3.7 **Resistance to thermal shock.**—The syringes shall not break (due to temperature changes) when submitted to the thermal shock test, as specified in 4.3.5.

3.8 **Annealing.**—Syringes shall be examined for strain, as specified in 4.3.6. A moderate amount of evenly distributed longitudinal strain is acceptable. Any ring strain or severe longitudinal strain shall not be acceptable.

3.9 **Permanency of marking.** — Syringes shall be subjected to the permanency of marking test as specified in 4.3.7. Syringes shall be considered to pass the test if not more than ½ of any graduation line or numeral has been lost, and the area showing complete pigment loss is less than 2 percent of the total pigmented area, and if the legibility of any portion of the markings is not seriously impaired.

3.10 **Accuracy.**—When tested as specified in 4.3.9, syringes shall deliver plus or minus 5 percent of the capacity indicated at each graduation mark tested.

3.11 **Workmanship.** — Workmanship shall be first class throughout. Syringes shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *For visual examination.* — Sampling procedures for visual examination shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

4.1.2 *For tests.*

4.1.2.1 *For alkalinity and density tests.*—Sample size for each test shall be as indicated for inspection level L–1 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero (0) for all sample sizes, and a minimum of two (2) samples shall be taken regardless of lot size. The unit of product for inspection purposes shall be one (1) syringe for density test, and the unit of product (or test unit) for alkalinity test shall consist of the number of syringes specified in table V.

4.1.2.2 *For other tests.*—Sample size for each test shall be as indicated for inspection level L–5 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero for all sample sizes. The unit of product for inspection purposes shall be one syringe.

4.2 **Inspection.**—The inspector shall ascertain that the syringes have been examined and tested in accordance with specification requirements.

4.3 **Tests.**

4.3.1 *Leakage test.*—The syringe shall be attached to testing machine having a water pressure gage. The pressure in pounds per square inch gage (p.s.i.g.) applied to different capacity syringes shall be as follows:

Capacity	Pressure (p.s.i.g.)
0.25 cc.	70
0.5 cc.	70
1.0 cc.	70
2.0 cc.	50
5.0 cc.	45
10.0 cc.	40
20.0 cc.	30
30.0 cc.	25
50.0 cc.	25
100.0 cc.	20

Insulin syringes:
Long type—70 p.s.i.g.
Short type—50 p.s.i.g.

4.3.1.1 *Leakage between plunger and barrel.*—Pressure used shall be as specified in 4.3.1. The water used as a testing fluid shall be let in through the tip of the syringe. The plunger shall be drawn to a point ¼ the distance of the graduations where it shall be allowed to remain for 10 seconds. The plunger shall then be drawn back to midway between first position and the top graduation mark where it shall be allowed to remain for a similar length of time. The plunger shall then be drawn back to the top graduation mark where it shall be allowed to remain for a similar length of time. The syringe shall be tested in a horizontal position.

4.3.1.2 *Fluid tightness, Luer tip, to hypodermic needle hub.*—A gage, as shown in figure 8 of ASA Standard Z70.1–1955, except that it shall be constructed with a suitable fluid path, shall be attached to the tip of the syringe and the water let in through the gage until the pressure is raised to the value as specified in 4.3.1, with the plunger held stationary at the halfway point. The syringe shall be allowed to remain under pressure for 10 seconds.

4.3.1.3 *Fluid tightness, needle adapter to hypodermic needle hub.*—Test shall be conducted as specified in 4.3.1.2, except that gage shall be attached to the needle adapter and the adapter shall be properly mounted on a syringe.

4.3.2 *Alkalinity.*—Syringes shall be washed in distilled water. All surfaces shall be brushed to remove loose particles and thoroughly rinsed with the water. Unassembled syringes shall then be placed in an alkali-free resistant glass beaker and covered with a solution of freshly boiled distilled water to which has been added 0.2 cc. of United States Pharmacopoeia (U.S.P.) phenolphthalein test solution for each 100 cc. of water. The number of syringes to be used, the size of the beaker, and the volume of water shall be as specified in table V. Syringes shall be autoclaved at 15 p.s.i.g. steam pressure for 15

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:53Z
Check the source to verify that this is the current version before use.

minutes and then removed from the autoclave. After the solution has cooled, any pink color which may be observed, shall be neutralized (titrated) with 0.02N sulphuric acid.

4.3.3 *Taper of tips.*

4.3.3.1 *Taper of Luer tip for types I and III syringes, and for needle adapter.*—The ground portion of tips for types I and III syringes, and for the tip of needle adapters, shall be tested with the ring gage shown in figure 7 of ASA Standard Z70.1–1955. This gage shall be slipped onto the tip with light pressure. In addition, when this ring gage is fitted onto the tip, as specified above, any movement of the gage on the tip when the gage is pressed lightly from side to side in all directions shall be noted.

4.3.3.2 *Taper of Luer tip (glass) for type II syringe.*—The ground portion of tips for type II syringes shall be tested with a standard female taper gage having a dimension at its extreme distal end of 0.230 inch and increasing in diameter at the rate of 0.060 inch per inch of length. The gage shall measure exactly 0.312 inch in thickness. This gage shall be slipped onto the tip with light pressure. When the ring gage is fitted onto the

TABLE V.—*Test for alkalinity of syringes*

Type	Size	Capacity	Number syringes (unassembled)	Size of beaker	Volume of water
		cc.		cc.	cc.
	1	2	5	150	140
I	2	5	5	400	325
	3	10	5	600	475
	4	20	5	1,500	1,100
	1	30	5	2,000	1,475
II	2	50	5	4,000	2,000
	3	100	4	4,000	2,600
	1	0.25	5	250	200
	2	0.5	5	250	200
	3	1	5	400	350
III	4	1	5	150	150
	5	1	5	400	350
	6	1	5	400	350
	7	1	5	400	350

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

tip, as specified above, any movement of the gage on the tip when the gage is pressed lightly from side to side in all directions shall be noted.

4.3.4 *Resistance to autoclaving.*—Syringes (dry) with the plunger pushed completely in the barrel shall be subjected to autoclave sterilization for 15 minutes at 15 p.s.i.g. steam pressure.

4.3.5 *Resistance to thermal shock.*— Syringes shall be transferred suddenly from boiling water into ice water (3° to 5° centigrade). This shock test shall be repeated at least 10 times.

4.3.6 *Annealing.*—Syringes shall be examined for strain with a polariscope.

4.3.7 *Permanency of marking test.*—The syringes shall be completely immersed in an open beaker containing hydrochloric acid solution, as described below. Autoclave the beaker containing the hydrochloric acid and syringes at 15 p.s.i.g. and 250° Fahrenheit (F.) for two periods of 30 minutes each. The 30-minute period shall begin at the instant the temperature of the hydrochloric acid solution in the autoclave reaches 245° F., as measured by a thermocouple immersed in the liquid. At the end of the 30-minute period, remove the syringes, rinse with cold tap water, and dry by wiping with cellulose tissue. Repeat the autoclaving operation and drying process, and then examine for loss of pigmentation. The hydrochloric acid solution shall be as follows:

Hydrochloric acid, 35 – 38 percent, U.S.P. or American Chemical Society (ACS) quality—10.0 milliliters (ml.)

Distilled water, quantity sufficient to make (q.s.) 1000 cc.

4.3.8 *Density.*—Density of glass shall be determined by any standard method.

4.3.9 *Accuracy test.* — Syringes shall be tested (with water) for accuracy at each main graduation mark (see tables I, II, and III) with a calibrated burette.

4.4 **Supplier responsibilities for inspection.** —The supplier shall be required to perform examinations and tests set forth in this specification to substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have adequate test facilities or shall arrange for test facilities acceptable to the Government, to be used for executing specific prescribed tests. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government upon the Government's request, at any time, or from time to time, for a period of 2 years after the delivery of the supplies to which these records relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

(Levels of preservation, packaging and packing protection used are defined in Fed. Std. No. 102 (for civil agencies only).

5.1 Packaging.

5.1.1 *Unit of issue.*—One syringe of the stock number specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table VI shall be packaged in each unit, intermediate, and shipping container, as applicable, for the required level of protection specified in procurement document.

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in a shipping container, such units shall be packed in a shipping container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

TABLE VI.

Stock No.	Type	Unit package (units)	Intermediate package (units)	Shipping container (units)
6515-282-9600	III	1	12	360
6515-380-2700	III	1	12	360
6515-380-4100	I	1	12	288
6515-380-4300	I	1	6	288
6515-380-4500	II	1	6	216
6515-514-2395	I	1	12	360
6515-598-5809	III	1	12	360
6515-598-5810	III	1	12	360
6535-812-2000	III	1	12	480

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–676 or Military Specification MIL–B–4229. Carton design shall include suitable interior fittings to insure against shifting or breakage. Each carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size and design. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the products for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**

5.2.1 *Level A.*

5.2.1.1 *Shipping container.*—The number of intermediate cartons specified in table VI shall be overpacked in an exterior container and constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specifications.

Note.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Shipping container.*—The number of intermediate cartons specified in table VI shall be overpacked in an exterior container constructed in accordance with Federal Specification PPP–B–621, table II.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Interior packages.*—Each unit package and each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 **Ordering data.** — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

(*a*) Title, number and date of this specification.

(*b*) Type and size required (see 1.2).

(*c*) If needle adapter for type II is not required (see 3.2 and 3.3.2.1).

(*d*) Design of tip, if other than specified (see 3.2.1.2).

(*e*) Graduation scale, if other than specified (see 3.3.1.1).

(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

6515–282–9600 SYRINGE, LUER, Tuberculin, 1 cc.
6515–380–2700 SYRINGE, LUER, Tuberculin, ¼ cc.
6515–380–4100 SYRINGE, LUER, 10 cc.
6515–380–4300 SYRINGE, LUER, 20 cc.
6515–380–4500 SYRINGE, LUER, 30 cc.
6515–514–2395 SYRINGE, LUER, 2 cc.
6515–598–5809 SYRINGE, LUER, Insulin, 40 units, 1 cc.
6515–598–5810 SYRINGE, LUER, Insulin, 80 units, 1 cc.
6535–812–2000 SYRINGE, LUER, Mallein, ½ cc.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Goverment requirements.

6.4 **Transportation description.** — The transportation description applicable to these items is:

Glassware, NOIBN (specify valuation)
Carload minimum weight 24,000 pounds subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 24,000 pounds subject to Rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.